Graham's Rearrangement Conjecture

Sequenceable Groups and Rainbow Paths in Cayley Digraphs

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Recap: Groups and Their Tables

Definition 1

A **group** $(G, \cdot)$: an associative operation with identity e and inverses. It is **abelian** if $xy = yx$; then we write $+$ and 0.

In the **Cayley table**, every row and every column contains each element once: $xy = xy' \Rightarrow y = y'$. So the table is a **Latin square**.

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Figure: The Cayley table of $\mathbb{Z}_4$.

Recap: Involutions in Abelian Groups

Definition 2

An **involution** is an element $g \neq e$ with $g^2 = e$.

Example 3

$\mathbb{Z}_4$ has one involution (2); $\mathbb{Z}_2^2$ has three; $\mathbb{Z}_n$ with n odd has none.

Every finite abelian group is $\mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}$; it has **exactly one** involution iff its Sylow 2-subgroup is cyclic and non-trivial.

From Group Operations to a Coloured Walk

An ordering $s_1, \ldots, s_k$ gives partial products $p_0 = e$, $p_i = s_1 \cdots s_i$.

$$p_{i-1} \xrightarrow{\text{colour } s_i} p_i = p_{i-1} s_i.$$

group language	**graph language**
the element s_i	the colour of step i
the partial product p_i	the vertex reached at step i
distinct terms s_i	distinct arc colours (rainbow)
distinct partial products	no repeated visited vertex

We will distinguish whether the initial vertex p_0 must also be distinct. That choice changes the required path length by one (Bucić et al., §1).

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Lucas's Round Dances, 1883

Problem 4 (Lucas, 1883: "Les rondes enfantines")

*Children dance in a ring, holding hands. How should the children be placed in successive rounds so that each child is, in turn, a **neighbour of every other child** — on the right or on the left — but **only once**?*

The solutions Lucas prints are due to **Walecki**, who taught at the Lycée Condorcet.

A necessary condition (Lucas). In every round a child has two neighbours, and no pair may meet twice, so each child meets the others two at a time: the number of *other* children is even. Hence there are $2n+1$ children, and there must be exactly n rounds.

In graph language: decompose K_{2n+1} into n Hamilton cycles.

Walecki's Solution for Eleven Children

Lucas's five rounds (child `A` stands in the centre of the figure):

I	A	B	C	D	E	F	G	H	I	J	K	A
II	A	C	E	B	G	D	I	F	K	H	J	A
III	A	E	G	C	I	B	K	D	J	F	H	A
IV	A	G	I	E	K	C	J	B	H	D	F	A
V	A	I	K	G	J	E	H	C	F	B	D	A

All $55 = \binom{11}{2}$ pairs of children are neighbours exactly once (checked).

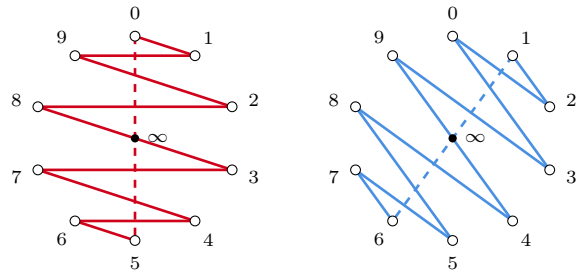


Figure: Walecki's "needle": put one child at the centre (∞) and 10 on a circle; the zigzag plus the two dashed edges is one round. Rotating the needle by one step gives the next round.

Where the Group Is Hiding

Label the $2n$ outer children by $\mathbb{Z}_{2n}$. Walecki's zigzag is

$$a = (0,\ 1,\ 2n-1,\ 2,\ 2n-2,\ 3,\ \ldots,\ n) \qquad (\text{for } 2n = 10:\ 0, 1, 9, 2, 8, 3, 7, 4, 6, 5).$$

Its successive **differences** are $1, -2, 3, -4, \ldots$, i.e.

$$b = (1,\ 2n-2,\ 3,\ 2n-4,\ \ldots) \qquad (\text{for } 2n = 10:\ 1, 8, 3, 6, 5, 4, 7, 2, 9),$$

every non-zero element of $\mathbb{Z}_{2n}$ exactly once.

Fact 5 (checked for $2n = 10$; Ollis, Example 4)

*The n rotations $a + i$ ($0 \le i < n$) decompose K_{2n} into Hamilton paths; joining the ends of each to ∞ gives Lucas's n rounds. All $2n$ rotations, read as directed paths, decompose the **complete digraph** on $\mathbb{Z}_{2n}$ into $2n$ directed Hamilton paths.*

Williams, 1949: Treatments Applied in Succession

The setting (Williams, 1949). n treatments (diets) are given **in succession** to the same units (cows): unit i receives all n treatments, one per period. Rows = units, columns = periods: a Latin square.

The difficulty. The response in a period depends on the treatment given (**direct effect**) and also on the treatment of the *previous* period (**residual** or **carry-over effect**):

$$y = \mu + \text{unit} + \text{period} + \tau_{\text{now}} + \rho_{\text{before}} + \varepsilon.$$

Balance. To estimate all ρ's equally well, every treatment should be **preceded by every other treatment equally often**.

Williams's example: diet and the milk yield of cows (Ollis, §3.6).

0	3	1	2
1	0	2	3
3	2	0	1
2	1	3	0

Figure: 4 cows, 4 periods, 4 diets: each diet follows each other diet exactly once.

Williams's Designs: Even and Odd n

Theorem 6 (Williams, 1949, abstract)

*A design balanced for the effect of the preceding treatment exists with n replications (**one** $n \times n$ square) when n is even, and with $2n$ replications (**two** squares) when n is odd.*

0	1	2
1	2	0
2	0	1

2	1	0
0	2	1
1	0	2

Figure: $n = 3$: the second square reverses the rows of the first. Together, each ordered pair is adjacent **twice**.

For $n = 3$ one square cannot work: none of the 3×3 Latin squares has all six ordered pairs adjacent in its rows.

Question: for which odd n does one square suffice?

Row-Complete Latin Squares

Definition 7 (Ollis, §1)

A Latin square is **row-complete** if every ordered pair (x, y), $x \neq y$, occurs **exactly once** in horizontally adjacent cells; it is **complete** if it is also column-complete.

Three faces of the same object, for n symbols:

$\triangleright$ a Williams design for n treatments using **one** square;

$\triangleright$ a row-complete Latin square of order n;

$\triangleright$ a decomposition of the complete digraph on n vertices into n arc-disjoint directed Hamilton paths (rows read as paths).

Row-Complete Squares Are Hamilton Path Decompositions

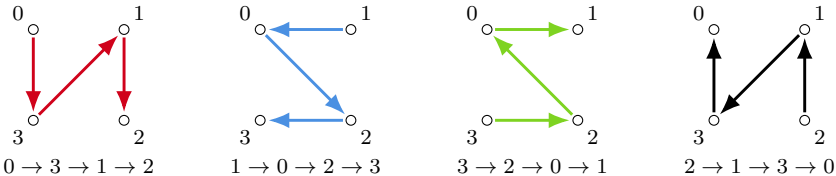


Figure: The rows of the 4×4 square as paths. The 12 arcs are distinct, so they are all the arcs of the complete digraph on 4 vertices.

Fact 8 (Ollis, §1)

A row-complete Latin square of order n decomposes the complete digraph on n vertices into n arc-disjoint Hamilton paths. (Each row visits every vertex once; each ordered pair is used once.)

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Gordon's Question, 1961

Problem 9 (Gordon, 1961)

"In an investigation concerning a certain type of Latin square, the following problem arose: Can the elements of a finite group G be arranged in a sequence $a_1, a_2, \ldots, a_n$ so that the partial products $a_1,\ a_1a_2,\ \ldots,\ a_1a_2\cdots a_n$ are all distinct?"

Definition 10 (Gordon, 1961)

Such an ordering $b_1, \ldots, b_n$ of G is a **sequencing**; G is **sequenceable** if it has one. Necessarily $b_1 = e$: if $b_i = e$ for some $i > 1$, then $b_1 \cdots b_{i-1} = b_1 \cdots b_i$.

Walecki's differences $(0, 1, 2n-2, 3, \ldots)$ are a sequencing of $\mathbb{Z}_{2n}$: its partial sums are the zigzag.

From a Sequencing to a Complete Latin Square

Theorem 11 (Gordon, 1961)

If $b_1, \ldots, b_n$ is a sequencing of G with partial products $a_i = b_1 \cdots b_i$, then $L_{ij} = a_i^{-1} a_j$ is a ***complete*** *Latin square.*

Why row-complete. If (x, y) sits in row i, columns $j, j+1$, then $x^{-1}y = a_j^{-1} a_{j+1} = b_{j+1}$. Distinct b's fix j; then $a_i = a_j x^{-1}$ fixes i.

Example 12 ($\mathbb{Z}_4$)

$b = (0, 3, 2, 1)$, $a = (0, 3, 1, 2)$.

$a_j - a_i$	0	3	1	2
0	0	3	1	2
3	1	0	2	3
1	3	2	0	1
2	2	1	3	0

Figure: The cows' square of the Williams slide.

What Gordon's Theorem Gives the Experimenter

A sequenceable group of order $n \Longrightarrow$ a one-square Williams design for n treatments.

▷ n even: $\mathbb{Z}_n$ is sequenceable (Lucas–Walecki), recovering Williams's even case.

▷ n odd: no abelian group of odd order is sequenceable (next slides), so one needs a **non-abelian** group of odd order.

▷ The non-abelian group of order 21 is sequenceable (Dénes–Tőrők; Mendelsohn, 1968): one 21×21 square balances 21 treatments.

So the experimental question became a question about groups.

Which Groups Are Sequenceable? Small Cases

Example 13 (sequenceable)

$\mathbb{Z}_4$: $(0, 3, 2, 1)$
$\mathbb{Z}_{2n}$: $(0, 1, 2n-2, 3, 2n-4, 5, \ldots, 2, 2n-1)$
D_{10} (Gordon)

Example 14 (not sequenceable)

$\mathbb{Z}_3$, $\mathbb{Z}_5$, $\mathbb{Z}_2^2$
D_6, D_8, Q_8 (Gordon)

What separates the two columns?

Trying It by Hand: Partial Sums as a Walk

Put the elements of the group on a circle. A sequencing $0, b_2, \ldots, b_n$ is a walk from 0 that takes **each non-zero element once as a step** and **never revisits a point**.

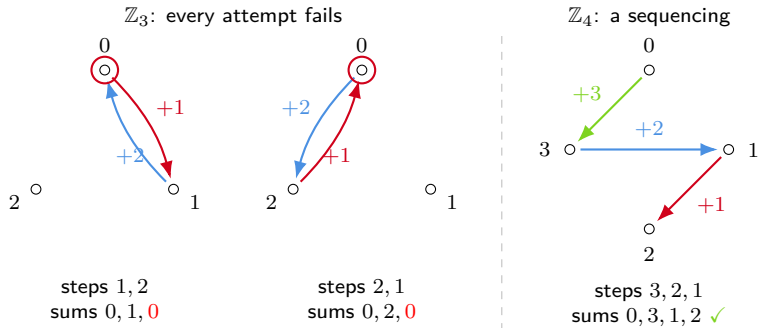


Figure: In $\mathbb{Z}_3$ both walks return to 0 (circled): the steps $1 + 2 = 3 \equiv 0$ always bring us back. In $\mathbb{Z}_4$ the steps $3, 2, 1$ visit every point once.

Abelian Groups: Gordon's Answer

Theorem 15 (Gordon, 1961)

A finite abelian group is sequenceable $\iff$ *it has* ***exactly one*** *involution.*

The obstruction. In an abelian group the last partial sum is $\sum_{g \in G} g$, whatever the order. Pairing g with $-g$, this is the sum of the involutions, which is 0 unless there is exactly one (Ollis, Thm 3). But $a_n = a_1 = 0$ is forbidden.

The answer is read off the group structure.

Non-Abelian Groups

▷ (Gordon, 1961) D_6, D_8, Q_8 are not sequenceable; D_{10} is.

▷ (1968–1973) D_{12}, D_{14}, D_{16}; the non-abelian groups of order 21, 27, 39, 55, 57 (Ollis, §2).

▷ (Anderson; Ollis) every non-abelian group of order $10 \leq n \leq 255$, by computer.

▷ Dihedral groups of order ≥ 10; soluble groups with one involution except Q_8 (Ollis, §2.6).

Theorem 16 (Müyesser–Pokrovskiy)

Every sufficiently large non-abelian group is sequenceable.

The only non-sequenceable groups known: abelian groups without a unique involution, and D_6, D_8, Q_8 (Ollis, §2.6).

Ringel's Variant: R-Sequencings

Definition 17 (Ringel, 1974)

An **R-sequencing** of G is an ordering $e = b_1, \ldots, b_n$ with $b_1 \cdots b_n = e$ and $b_1,\ b_1 b_2, \ldots,\ b_1 \cdots b_{n-1}$ distinct.

Ringel was led to it by constructions in his proof of the Heawood map-colour theorem (Bucić et al., §1).

Example 18

$\mathbb{Z}_3$: $(0, 1, 2)$; $\mathbb{Z}_5$: $(0, 1, 2, 4, 3)$, partial sums $0, 1, 3, 2, 0$.

Fact 19 (Paige, 1951; Ollis, §3.1)

*An R-sequencing gives a **complete mapping**, hence a pair of orthogonal Latin squares — the objects of deck 01.*

R-Sequencings and the Hall–Paige Condition

For abelian groups, sequenceable and R-sequenceable exclude each other: the last partial sum is fixed (Ollis, §3.1).

Theorem 20 (Friedlander–Gordon–Miller, 1978; proved by Alspach–Kreher–Pastine, 2017)

An abelian group is R-sequenceable $\iff$ *its Sylow* 2*-subgroup is trivial or non-cyclic.*

▷ (Müyesser–Pokrovskiy) a large group is R-sequenceable $\iff$ it satisfies the Hall–Paige condition.

▷ (Pastine–Soldera Ruiz, 2026) every group of order coprime to 30.

The same condition as Hall–Paige.

R-Sequencings: Further Families in 2026

Theorem 21 (Pastine–Soldera Ruiz, 2026, Cor. 2.7–2.8)

Each of the following groups is R-sequenceable:

- *groups whose order is coprime to* 30*;*
- *nilpotent groups whose order is coprime to* 6*, except possibly groups of order* 5^k*.*

Here a finite nilpotent group is a direct product of its Sylow subgroups. The paper proves the stronger property of R**-sequenceability.

These are whole-group results: they do not assert that every subset has a valid ordering. The possible 5-group exception is retained.

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Colouring a Cayley Digraph

Definition 22

For $S \subseteq G \setminus \{e\}$, the **Cayley digraph** $\mathrm{Cay}(G, S)$ has vertex set G and an arc $g \to gs$ of **colour** s for each $g \in G$, $s \in S$. A directed path or cycle is **rainbow** if its arcs have distinct colours.

Lemma 23

1. *The colouring is **proper**: at every vertex there is exactly one out-arc and one in-arc of each colour.*
2. *For $h \in G$, the map $x \mapsto hx$ sends the arc $g \to gs$ to the arc $hg \to hgs$ of the **same colour**.*

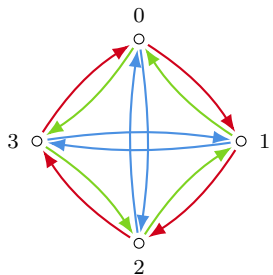


Figure: $\mathrm{Cay}(\mathbb{Z}_4, \{1, 2, 3\})$; colours 1, 2, 3.

The Problems, Restated

For an ordering $s_1, \ldots, s_k$ of $S \subseteq G \setminus \{e\}$ write $p_0 = e$ and $p_t = s_1 \cdots s_t$.

Problem 24 (Gordon, translated)

*For which finite groups G does $\mathrm{Cay}(G, G \setminus \{e\})$ contain a **rainbow directed Hamilton path**?*

Problem 25 (Ringel, translated)

*For which G does it contain a **rainbow directed cycle** with $|G| - 1$ arcs?*

Problem 26 (Graham, translated)

*For a prime p and $S \subseteq \mathbb{Z}_p \setminus \{0\}$, does $\mathrm{Cay}(\mathbb{Z}_p, S)$ contain a **rainbow directed path** with $|S| - 1$ arcs?*

Each problem is *equivalent* to the original one — the following equivalences.

The Equivalence, Part 1: Whole Groups

Proposition 27

Let $|G| = n$. (a) G is sequenceable $\iff$ $\mathrm{Cay}(G, G \setminus \{e\})$ has a rainbow directed Hamilton path. (b) G is R-sequenceable $\iff$ it has a rainbow directed cycle with $n-1$ arcs.

Proof of (a). $\Rightarrow$: for a sequencing $e = b_1, b_2, \ldots, b_n$ the partial products $a_1 = e, a_2, \ldots, a_n$ are the n vertices, and the arc $a_t \to a_{t+1} = a_t b_{t+1}$ has colour b_{t+1}; the colours $b_2, \ldots, b_n$ are distinct. $\Leftarrow$: let $v_1 \to \cdots \to v_n$ be a rainbow Hamilton path, with $v_{t+1} = v_t c_{t+1}$. By the Lemma, $u_t = v_1^{-1} v_t$ is again a rainbow Hamilton path, now starting at $u_1 = e$. Then $(e, c_2, \ldots, c_n)$ is a sequencing: its partial products are $u_1, \ldots, u_n$.

(b) is the same argument: the cycle $e \to b_2 \to b_2 b_3 \to \cdots \to e$ uses every colour once. □

The Equivalence, Part 2: Subsets

Call an ordering of S **valid** if $p_1, \ldots, p_k$ are distinct, and **strongly valid** if $p_0, p_1, \ldots, p_k$ are distinct.

Proposition 28

Let $|S| = k$. In $\mathrm{Cay}(G, S)$:

(a) *A valid ordering $\iff$ a rainbow directed path with $k-1$ arcs.*

(b) *A strongly valid ordering $\iff$ a rainbow directed path with k arcs.*

(c) *An ordering with $p_0, \ldots, p_{k-1}$ distinct and $p_k = e$ $\iff$ a rainbow directed cycle with k arcs.*

In (a), the path uses all but one colour. In (b) and (c), it uses every colour.

Why One Missing Colour Is Enough

Suppose $|S| = k$ and a rainbow path $v_1 \to \cdots \to v_k$ uses colours $c_2, \ldots, c_k$. Let $c \in S$ be the unused colour.

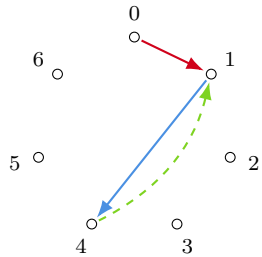
1. Translate every vertex by left multiplication with $h = cv_1^{-1}$. The first vertex becomes $hv_1 = c$, and all arc colours stay the same.
2. Read the translated vertices as the partial products of $(c, c_2, \ldots, c_k)$. They are distinct because the path is simple.
3. Conversely, distinct partial products $p_1, \ldots, p_k$ give the rainbow path $p_1 \to \cdots \to p_k$.

For a strongly valid ordering, include $p_0 = e$ to obtain k arcs. If only $p_k = p_0$ repeats, the walk is a cycle.

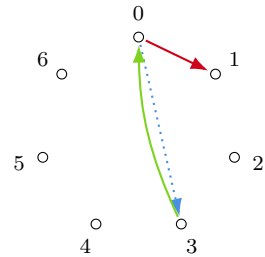
This is the equivalence behind Graham's rainbow-path formulation (Bucić et al., §1).

An Example in $\mathbb{Z}_7$

Take $S = \{1, 3, 4\} \subseteq \mathbb{Z}_7$.



order $1, 3, 4$: sums $1, 4, 1$



order $3, 4, 1$: sums $3, 0, 1$

Figure: Arc $x \to x + s$ has colour 1, 3 or 4. Left: the walk from 0 revisits 1. Right: the partial sums $3 \to 0 \to 1$ form a rainbow path (the dotted arc $0 \to 3$ is the first step $s_1 = 3$). Four of the six orders of S work.

The Dictionary

algebra	**combinatorics**
group G, subset S	$\mathrm{Cay}(G, S)$, arc $g \to gs$ coloured s
left multiplication	colour-preserving automorphism
valid ordering of S	rainbow path with $\lvert S\rvert - 1$ arcs
strongly valid ordering of S	rainbow path with $\lvert S\rvert$ arcs
sequencing of G	rainbow Hamilton path in $\mathrm{Cay}(G, G \setminus \{e\})$
R-sequencing of G	rainbow cycle with $\lvert G\rvert - 1$ arcs
sequencing $\Rightarrow$ complete Latin square	Hamilton path decomposition of the complete digraph
complete mapping (deck 01)	rainbow perfect matching in $K_{n,n}$

R-Sequencings as Cyclic Vertex Orders

We can now return to the whole-group R-sequencing question.

Definition 29 (Javaheri, 2026, Def. 1.1)

An equivalent convention lists the nonidentity elements as $g_1, \ldots, g_{\ell-1}$, with consecutive quotients

$$g_1^{-1} g_2, \ldots, g_{\ell-2}^{-1} g_{\ell-1}, g_{\ell-1}^{-1} g_1$$

also listing $G \setminus \{e\}$. It is an R*-sequencing if, for some cyclic index i, $g_{i-1} g_{i+1} = g_{i+1} g_{i-1} = g_i$.

Here the g_i are **vertices**; the quotients are **arc colours**. Thus the cycle uses every nonidentity vertex and every nonidentity colour.

To compare with the earlier definition, translate an R-cycle so its missing vertex is e. Translation preserves the quotient colours.

A Direct-Product Extension in 2026

The extra R* condition allows a construction on direct products.

Theorem 30 (Javaheri, 2026, Thm. 1.3)

If G has even order and is R-sequenceable, and H has odd order, then $G \times H$ is R*-sequenceable.*

▷ In the product group, this gives a rainbow cycle of length $|G||H| - 1$, using all nonidentity quotient colours.

▷ The input about H is a cyclic ordering whose adjacent products enumerate H and whose positions respect inversion: a **symmetric harmonious ordering** (deck 03).

This concludes the whole-group extension. We next return to arbitrary subsets and the rainbow-path question. Preprint: November 2025; journal publication: 2026.

Why the Graph Language Helps

$\mathrm{Cay}(G, S)$ is a $|S|$-regular digraph with a proper $|S|$-colouring of its arcs. So any theorem about **all** such digraphs applies to every group and every S at once.

Problem 31 (Schrijver's question; directed version by Bucić et al.)

Does every d-regular (di)graph whose edges are properly coloured with d colours contain a rainbow (directed) path with $d-1$ edges?

A positive answer to the directed version gives, by Part 2(a), a valid ordering of **every** $S \subseteq G \setminus \{e\}$ in **every** group. The recent progress on Graham's conjecture runs through such rainbow-path theorems (Bucić et al., 2025).

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Graham's Question, 1971

Conjecture 1 (Graham, 1971; repeated by Erdős–Graham, 1980)

For every prime p, every $S \subseteq \mathbb{Z}_p \setminus \{0\}$ has a ***valid ordering****: an ordering $s_1, \ldots, s_k$ whose partial sums $s_1,\ s_1 + s_2,\ \ldots,\ s_1 + \cdots + s_k$ are distinct.*

▷ Posed in a paper on sums of integers taken from a fixed sequence, among problems on sumsets; partly motivated by **juggling**, according to later papers of Graham (Bucić et al., §1); (Buhler et al., 1994).

▷ The same question appeared **independently** twice: from **Heffter arrays** (Archdeacon et al., 2015) and from **cycle decompositions** (Alspach). Next five slides.

In the dictionary: every $\mathrm{Cay}(\mathbb{Z}_p, S)$ has a rainbow path with $|S| - 1$ arcs.

Heffter Arrays

Definition 32 (Archdeacon, 2015; Pasotti–Dinitz, Def. 1.4)

A **Heffter array** $H(m, n; h, k)$ is an $m \times n$ partially filled array with entries in $\mathbb{Z}_v$, $v = 2nk + 1$, such that

1. each row has h filled cells and each column has k;
2. for every $x \in \mathbb{Z}_v \setminus \{0\}$, exactly one of $x, -x$ appears;
3. every row and every column sums to 0 in $\mathbb{Z}_v$.

6	7	-10	-4	1
-9	5	2	-11	13
3	-12	8	15	-14

Figure: An $H(3, 5)$ over $\mathbb{Z}_{31}$ (Pasotti–Dinitz, Example 1.3).

Entries: $\pm 1, \ldots, \pm 15$, each up to sign once. Row sums: $6 + 7 - 10 - 4 + 1 = 0$, …; column sums: $6 - 9 + 3 = 0$, … (checked).

From a Heffter Array to Cycle Decompositions

An ordering $t_1, \dots, t_h$ of a row is **simple** if its partial sums are distinct (Pasotti–Dinitz, §2); since the row sums to 0, the last one is 0.

Row $\to$ cycle. The partial sums $0, t_1, t_1 + t_2, \dots$ form an h-cycle in K_v whose edges have "lengths" $\pm t_1, \dots, \pm t_h$. The rows use every length $\pm x$ exactly once, so the cycles of all rows and all their translates $+g$ ($g \in \mathbb{Z}_v$) use every edge of K_v exactly once.

Fact 33 (Archdeacon, 2015; Pasotti–Dinitz, §5)

If every row and column of $H(m, n; h, k)$ has a simple ordering, then K_v has a cyclic decomposition into h-cycles (from the rows) and one into k-cycles (from the columns), and the two are orthogonal.

Example 34 (the $H(3, 5)$, checked)

Row $6, 7, -10, -4, 1$: sums $6, 13, 3, 30, 0$ — simple. Rows: 93 five-cycles decomposing K_{31}. Columns: 155 triangles — a Steiner triple system of order 31.

The Archdeacon–Dinitz–Mattern–Stinson Conjecture

To use *any* Heffter array one needs a simple ordering of *every* row. Archdeacon, Dinitz, Mattern and Stinson asked for it in general:

Conjecture 2 (Archdeacon–Dinitz–Mattern–Stinson, 2015)

For every n, every $A \subseteq \mathbb{Z}_n \setminus \{0\}$ has a valid ordering.

Example 35 (their Example 1.1)

$A = \{1, \ldots, 6\} \subseteq \mathbb{Z}_8$, order $1, 6, 3, 4, 5, 2$: partial sums $1, 7, 2, 6, 3, 5$.

This is Graham's conjecture with $\mathbb{Z}_p$ replaced by $\mathbb{Z}_n$ "(although their work was independent of Graham)" (Ollis, §3.7).

Alspach's Cycle Decomposition Problem

Problem 36 (Alspach, 1981)

*For odd n, can K_n be decomposed into cycles of **any** prescribed lengths $m_1, \ldots, m_t$ with $3 \le m_i \le n$ and $\sum m_i = \binom{n}{2}$? (For even n: K_n minus a perfect matching.)*

Theorem 37 (Bryant–Horsley–Pettersson, 2014)

Yes: the obvious necessary conditions are sufficient.

Cycle decompositions of K_n and $K_n \pm I$, and directed cycle decompositions of complete digraphs, were Alspach's motivation for the following partial-sums question (Hicks–Ollis–Schmitt, §1, citing Bode–Harborth). The bridge is the next slide.

Partial Sums Build Cyclic Decompositions

Write $C_n(A)$ for the circulant graph $\mathrm{Cay}(\mathbb{Z}_n, A \cup -A)$.

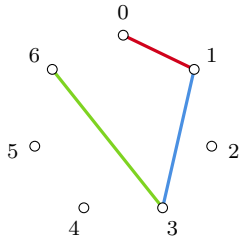
Proposition 38 (the difference method; cf. Pasotti–Dinitz, Prop. 5.5)

Let $A \subseteq \mathbb{Z}_n \setminus \{0\}$, $|A| = k$, $A \cap (-A) = \emptyset$, and let $s_0 = 0, s_1, \ldots, s_k$ be the partial sums of an ordering of A.

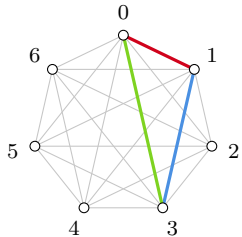
1. *If $s_0, \ldots, s_k$ are distinct, the n translates of the path $s_0 s_1 \cdots s_k$ decompose $C_n(A)$ into paths with k edges.*
2. *If $s_k = 0$ and $s_0, \ldots, s_{k-1}$ are distinct, the n translates of the cycle $s_0 s_1 \cdots s_{k-1}$ decompose $C_n(A)$ into k-cycles.*

Why. The edge $\{x, x+a\}$ with $a = a_t$ lies in exactly one translate, namely the one by $g = x - s_{t-1}$; $A \cap (-A) = \emptyset$ makes a (hence t) unique. Counting: $C_n(A)$ has nk edges, n blocks of k edges.

Two Examples in $\mathbb{Z}_7$



$A = \{1, 2, 3\}$: path $0, 1, 3, 6$
7 translates $= K_7$



$A = \{1, 2, 4\}$: triangle $0, 1, 3$
7 translates = Fano plane

Figure: Left: $1, 2, 3$ has partial sums $0, 1, 3, 6$; since $\pm\{1, 2, 3\}$ is all of $\mathbb{Z}_7 \setminus \{0\}$, the 7 translates of the path decompose K_7. Right: $1, 2, 4$ has partial sums $0, 1, 3, 0$; the 7 translates of the triangle $\{0, 1, 3\}$ are the lines of the Fano plane, a Steiner triple system of order 7 (checked).

Alspach's Conjecture and the Whole Family

Conjecture 3 (Alspach; see Bode–Harborth, 2005)

Every $A \subseteq \mathbb{Z}_n \setminus \{0\}$ with $\sum_{a \in A} a \neq 0$ has a ***strongly valid*** *ordering: $0, s_1, \ldots, s_k$ are distinct.*

Alspach's conjecture implies the ADMS conjecture (ADMS, Prop. 1.1).

conjecture	**group**	**in the dictionary**
Graham, 1971	$\mathbb{Z}_p$	rainbow path, $\lvert S\rvert - 1$ arcs
Archdeacon et al., 2015	$\mathbb{Z}_n$	rainbow path, $\lvert S\rvert - 1$ arcs
Alspach	$\mathbb{Z}_n$, $\Sigma A \neq 0$	rainbow path, $\lvert S\rvert$ arcs
Ollis, Conj. 38	abelian	path with $\lvert S\rvert$ arcs, or cycle
Bedert et al., Q. 1.1	any group	rainbow path, $\lvert S\rvert - 1$ arcs

Progress on Graham's Conjecture: Small Sets

▷ (Hicks–Ollis–Schmitt, 2019) $|S| \leq 10$ and $|S| \geq p - 3$.

▷ (Costa–Della Fiore–Ollis–Rovner-Frydman, 2022) $|S| \leq 12$.

▷ Sawin (2015); (Kravitz, 2024) $|S| \leq \log p / \log \log p$.

▷ (Bedert–Kravitz, 2024) $|S| \leq e^{c(\log p)^{1/4}}$ — beyond the "rectification barrier".

Methods: the polynomial method; rectification; dissociated sets.

Progress on Graham's Conjecture: Large Sets

▷ (Müyesser–Pokrovskiy) $|S| \geq (1 - o(1))p$.

▷ (Bucić–Frederickson–Müyesser–Pokrovskiy–Yepremyan, 2025) any group, any S: an ordering with all but $o(|S|)$ partial products distinct.

▷ (Bedert–Bucić–Kravitz–Montgomery–Müyesser, 2025) any group: every S with $|S| \geq |G|^{1-c}$.

Methods: rainbow paths in properly coloured digraphs; absorption; Freiman–Ruzsa.

Graham's Conjecture for Large Primes

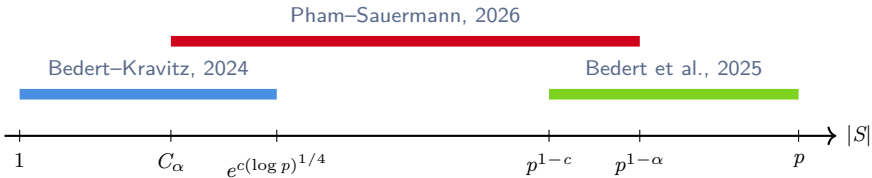


Figure: The three ranges overlap (not to scale).

Theorem 39 (Pham–Sauermann, 2026, with the earlier results)

Graham's rearrangement conjecture holds for all sufficiently large primes p.

In graph language, every $\mathrm{Cay}(\mathbb{Z}_p, S)$ has a rainbow directed path with $|S| - 1$ arcs.
Their range $C_\alpha \le |S| \le p^{1-\alpha}$ uses anticoncentration of sums of random subsets.

Why the Three Ranges Cover Every Subset

Theorem 40 (Pham–Sauermann, February 2026, Thm. 1.2)

For every $0 < \alpha < 1$ there is C_α such that every $S \subseteq \mathbb{Z}_p \setminus \{0\}$ with $C_\alpha \leq |S| \leq p^{1-\alpha}$ has a valid ordering.

Choose $0 < \alpha < c$, where $c > 0$ is from the large-set theorem.

▷ $|S| < C_\alpha$: the earlier small-set bound applies once p is large (Bedert–Kravitz).

▷ $C_\alpha \leq |S| \leq p^{1-\alpha}$: the new theorem.

▷ $|S| \geq p^{1-c}$: Bedert et al.

Since $p^{1-c} < p^{1-\alpha}$, there is no gap. The conclusion is for **all sufficiently large primes**; no explicit small-prime cutoff is asserted here.

Small Subsets of General Cyclic Groups

Theorem 41 (Costa–Della Fiore, February 2026, Thm. 1.3)

There is an absolute $c > 0$ such that, if p is the least prime divisor of k, every nonempty $A \subseteq \mathbb{Z}_k \setminus \{0\}$ with

$$|A| \leq \exp\bigl(c(\log p)^{1/3}\bigr)$$

is sequenceable: it has distinct partial sums and no zero proper partial sum.

This improves the earlier exponent $1/4$ to $1/3$ for general cyclic groups. The prime case above controls every subset. For general cyclic groups, this theorem controls the subset size using the least prime divisor.

The Small-Set Theorem Gives a Path or a Cycle

Under the preceding size bound, order $A = \{a_1, \ldots, a_r\}$ and put $p_0 = 0$, $p_i = a_1 + \cdots + a_i$ in $\mathbb{Z}_k$.

The stronger conclusion (Costa–Della Fiore, Thm. 1.3)

$p_0, \ldots, p_{r-1}$ are distinct, and the only allowed return is $p_r = p_0$.

▷ If $\sum A \neq 0$, the r steps give a rainbow directed path.

▷ If $\sum A = 0$, they give a rainbow directed cycle.

▷ Both use **every colour in** A. A valid ordering alone only asks for a path with $r - 1$ arcs.

Directed 2-cycles are allowed. The local t-weak variant (Thm. 1.4) is a separate conclusion, not a global ordering theorem for arbitrary $|A|$.

Beyond $\mathbb{Z}_p$: Composite Moduli

Theorem 42 (Costa–Della Fiore–Feng–Liu, August 2026, Thm. 1.3)

Fix $L \geq 2$ and $\gamma \geq 1$. For a sufficiently large $p_0 = p_0(L, \gamma)$, every subset of $\mathbb{Z}_k \setminus \{0\}$ has a valid ordering if, for distinct primes p_i and positive integers e_i,

$$k = \prod_{i=1}^{s} p_i^{e_i}, \qquad \sum_{i=1}^{s} e_i \leq L, \qquad p_0 \leq p_1 < \cdots < p_s \leq \gamma p_1.$$

This covers prime powers of bounded exponent and bounded products of comparable large primes. The bound is on the **sum of exponents**. Every subset gives a rainbow path with one fewer arc than colours.

Composite Moduli with a Fixed Small Factor

The previous theorem requires every prime factor to be large. A separate corollary allows a fixed factor t.

Theorem 43 (Costa–Della Fiore–Feng–Liu, Cor. 1.6)

Fix $t \geq 2$ with $\mathbb{Z}_t$ strongly sequenceable. For all sufficiently large primes $p \nmid t$, every subset of $\mathbb{Z}_{tp} \setminus \{0\}$ has a valid ordering. In particular, every fixed $2 \leq t \leq 21$ is allowed.

Here strongly sequenceable means that every nonzero subset admits an ordering with all partial sums, including the initial 0, distinct, except that the final sum may return to 0.

Strong sequenceability asks for paths or cycles using every colour. The **conclusion on $\mathbb{Z}_{tp}$** guarantees paths with $|A| - 1$ arcs.

Beyond Cyclic Groups; a Link to Deck 06

▷ (Bedert–Bucić–Kravitz–Montgomery–Müyesser, 2025) in $\mathbb{F}_2^n$, every nonzero subset S with $|S| \geq C$ has a valid ordering.

▷ (Costa–Della Fiore–Engel, 2025) small-set results for $\mathbb{Z}_p \rtimes_\varphi H$, where H is finite abelian and strongly sequenceable and each $\varphi(h)$ acts as $+1$ or -1: every nonidentity subset with $|S| \leq \exp(c(\log p)^{1/4})$ has a sequencing.

▷ The weak arithmetic regularity lemma developed in the Graham programme is also used in the 2026 theorem on moderately dense Cayley graphs (Bedert–Draganić–Müyesser–Pavez-Signé, §1).

The same group structure helps organise rainbow paths and Hamilton cycles.

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1. Recap: Groups and Cayley Tables

2. Motivation I: Round Dances and Carry-Over Effects

3. Motivation II: Gordon's Question

4. The Translation: Rainbow Paths in Cayley Digraphs

5. Graham's Conjecture and Its Rediscoveries

6. Open Problems

Open Problem 1: Keedwell's Conjecture

Conjecture 4 (Keedwell, 1981; Ollis, Conj. 15)

Every finite non-abelian group other than D_6, D_8 and Q_8 is sequenceable.



Proposition 27(a)

Graph form

For every finite non-abelian group $G \notin \{D_6, D_8, Q_8\}$, the Cayley digraph $\mathrm{Cay}(G, G \setminus \{e\})$, with arc $g \to gs$ coloured s, contains a **rainbow directed Hamilton path**.

Known. all non-abelian groups of order 10 to 255; dihedral groups of order ≥ 10; soluble groups with a unique involution except Q_8; every sufficiently large non-abelian group (Müyesser–Pokrovskiy) (Ollis, §2.6).
Open: non-abelian groups of order ≥ 256 outside these families and below the (inexplicit) threshold — "10^{50} or so should be comfortably enough" (Müyesser, in Ollis, §2.5).

Open Problem 2: Graham's Conjecture for Every Prime

Conjecture 5 (Graham, 1971)

*For **every** prime p, every $S \subseteq \mathbb{Z}_p \setminus \{0\}$ has an ordering $s_1, \ldots, s_k$ with $s_1,\ s_1 + s_2,\ \ldots,\ s_1 + \cdots + s_k$ distinct.*

$\Updownarrow$ Proposition 28(a)

Graph form

For every prime p and every $S \subseteq \mathbb{Z}_p \setminus \{0\}$, the digraph $\mathrm{Cay}(\mathbb{Z}_p, S)$, with arc $x \to x + s$ coloured s, contains a **rainbow directed path with $|S| - 1$ arcs**.

Known. all $p \geq p_0$ (Pham–Sauermann, 2026, with earlier results); every p when $|S| \leq 12$ or $|S| \geq p - 3$ (HOS), (CDOR); $p \leq 23$, from the computer check of the ADMS conjecture for $n \leq 25$ (HOS, §1).
Open: primes $29 \leq p < p_0$ with $13 \leq |S| \leq p - 4$; no explicit p_0 is known.

Open Problem 3: The Archdeacon–Dinitz–Mattern–Stinson Conjecture

Conjecture 6 (Archdeacon–Dinitz–Mattern–Stinson, 2015)

For every n, every $A \subseteq \mathbb{Z}_n \setminus \{0\}$ has an ordering $a_1, \ldots, a_k$ whose partial sums $s_1, \ldots, s_k$ are distinct.

$\Updownarrow$ Proposition 28(a)

Graph form

For every n and every $A \subseteq \mathbb{Z}_n \setminus \{0\}$, $\mathrm{Cay}(\mathbb{Z}_n, A)$ contains a **rainbow directed path with $|A| - 1$ arcs**.

Known. $n \leq 25$ by computer; $|A| \leq 6$ (HOS, §1); $n = tp$ with $\mathbb{Z}_t$ strongly sequenceable and p a large prime, and n whose prime factors are large and comparable with bounded total prime multiplicity (CDFL, 2026).
Why it matters: a simple ordering of every row and column of any Heffter array, hence its cycle decompositions.

Open Problem 4: Alspach's Conjecture

Conjecture 7 (Alspach; see Bode–Harborth, 2005)

For every n, every $A \subseteq \mathbb{Z}_n \setminus \{0\}$ with $\sum_{a\in A} a \neq 0$ has an ordering whose partial sums $0, s_1, \ldots, s_k$ are distinct.

$\Updownarrow$ Proposition 28(b)

Graph form

For every n and every $A \subseteq \mathbb{Z}_n \setminus \{0\}$ with $\sum_{a\in A} a \neq 0$, $\mathrm{Cay}(\mathbb{Z}_n, A)$ contains a **rainbow directed path with $|A|$ arcs** (using every colour).

Known. $|A| \in \{n-1, n-2\}$; claimed for $|A| \leq 5$ and for $n \leq 16$ (Bode–Harborth); for prime n: $|A| \leq 10$ and $|A| \geq n-3$ (HOS); $|A| \leq \exp(c(\log p)^{1/3})$, where p is the least prime divisor of n (Costa–Della Fiore, 2026). It implies Open Problem 3 (Prop. 1.1).

Open Problem 5: Strongly Sequenceable Abelian Groups

Conjecture 8 (Ollis, Conj. 38; question of Alspach–Kalinowski, see Alspach–Liversidge)

In every finite abelian group G, every $S \subseteq G \setminus \{0\}$ has an ordering whose partial sums $0, s_1, \ldots, s_k$ are distinct, or are distinct except that $s_k = 0$.

$\Updownarrow$ Proposition 28(b),(c)

Graph form

For every finite abelian group G and every $S \subseteq G \setminus \{0\}$, $\mathrm{Cay}(G, S)$ contains a **rainbow directed path with $|S|$ arcs** or a **rainbow directed cycle with $|S|$ arcs**.

Known. $|S| \leq 9$ (Alspach–Liversidge, 2020); further small $|S|$ under conditions on the group (Ollis, Thm 39). It implies Open Problems 3 and 4 for cyclic groups.

Open Problem 6: All Groups

Bedert et al., Question 1.1

For which finite groups G does every $S \subseteq G \setminus \{e\}$ have an ordering $s_1, \ldots, s_k$ with $s_1,\ s_1 s_2,\ \ldots,\ s_1 \cdots s_k$ distinct?

$\Updownarrow$ Proposition 28(a)

Graph form

For which finite groups G does $\mathrm{Cay}(G, S)$ contain a **rainbow directed path with $|S| - 1$ arcs** for **every** $S \subseteq G \setminus \{e\}$?

Known. every S with $|S| \geq |G|^{1-c}$, in every group; $\mathbb{F}_2^n$ for all $|S| \geq C$ (Bedert et al.). "It seems feasible that the answer … is affirmative for every finite group" (ibid.).

Open Problem 7: Row-Complete Latin Squares of Prime Order

Ollis, §3.6

Is there a row-complete Latin square of prime order $p \geq 13$? (Equivalently, a Williams design for p treatments with a single square.)

$\Updownarrow$ rows = paths, columns Latin

Graph form

Can the complete digraph on p vertices be decomposed into p directed Hamilton paths $P_1, \ldots, P_p$ such that, for each j, the j-th vertices of $P_1, \ldots, P_p$ are all different?

Known. every even order; every odd composite order (Higham, 1998); none of order $3, 5, 7$, nor 11 (Best–Wanless, reported in Ollis, §3.6).
Why groups cannot help: the only group of order p is $\mathbb{Z}_p$, which has no involution and so is not sequenceable.

Open Problem 8: The Directed Schrijver Problem

Bucić et al., Problem 1.6

Let D be a d-regular digraph whose arcs are properly coloured with d colours. Does D contain a rainbow directed path with $d-1$ arcs?

$\Updownarrow$ specialise to $D = \mathrm{Cay}(G, S),\ d = |S|$

Algebraic consequence

A positive answer gives, in **every** finite group, an ordering of **every** $S \subseteq G \setminus \{e\}$ with distinct partial products — Open Problems 2, 3 and 6 at once (Bucić et al., §1).

Known. rainbow paths with $(1-o(1))d$ arcs when $d = \Omega(n)$ (Bucić et al., Thm 1.7); asymptotically for complete symmetric digraphs (Benzing–Pokrovskiy–Sudakov, 2020); undirected Schrijver question: $(1-o(1))d$ edges (Bucić et al., Thm 1.5).

Summary: Algebra and Combinatorics

▷ **Experiments and dances** (Lucas 1883, Williams 1949) ask for row-complete Latin squares.

▷ **Gordon** (1961): a sequenceable group gives one — which groups are sequenceable is a question about the group.

▷ **Graham** (1971), **Archdeacon et al.** (Heffter arrays), **Alspach** (cycle decompositions) ask the same for subsets.

▷ **The translation**: distinct partial products = a rainbow path in the Cayley digraph; the group is used only through the colour-preserving translations.

Distinct partial products = rainbow paths in a Cayley digraph.

Thanks !

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